

1. (a) We take the estimation of population mean μ as an example. But the notion of unbiasedness and efficiency is widely applied in all kinds of estimation problems.

Suppose that our first candidate for the estimator is the sample average $\bar{Y}$, and the second candidate is defined as $\tilde{Y} = \frac{1}{2}(Y_1 + Y_n)$ (average of the first and last observations).

We would like to compare the two candidates using the following two criteria: unbiasedness and efficiency. First, as discussed in class, the sample average is unbiased: $E[\bar{Y}] = \mu$, and $\tilde{Y}$ is also unbiased:

$$E[\tilde{Y}] = E\left[\frac{1}{2}(Y_1 + Y_n)\right] = \frac{1}{2}E[Y_1 + Y_n] = \frac{1}{2}\{E[Y_1] + E[Y_n]\} = \frac{1}{2}\{\mu + \mu\} = \mu. \quad (1)$$

Therefore, both $\bar{Y}$ and $\tilde{Y}$ satisfy the criterion of unbiasedness.

Then, we compare the two candidates by using efficiency. As discussed in class, $Var[\tilde{Y}] = \sigma^2/n$. On the other hand, we note that

$$Var[\tilde{Y}] = Var\left[\frac{1}{2}(Y_1 + Y_n)\right] = \frac{1}{4}Var[Y_1 + Y_n] = \frac{1}{4}\{Var[Y_1] + Var[Y_n]\} \quad (2)$$

$$= \frac{1}{4}\{\sigma^2 + \sigma^2\} = \frac{\sigma^2}{2}, \quad (3)$$

where the third equality follows from the independence assumption. Therefore, $\bar{Y}$ is more efficient (has a smaller variance) than $\tilde{Y}$ as long as $n > 2$.

In sum, the sample average is considered a good estimator according to this comparison because both are unbiased but in general the sample average will be more efficient.

- (b) Gauss-Markov Theorem:

Under the 3 OLS assumptions: 1) the conditional distribution of u_i given X_i has a mean of zero, 2) $(X_i, Y_i), i = 1, \dots, n$ are independently and identically distributed (i.i.d.), and 3) large outliers are unlikely (and some other technical assumptions), **the OLS estimators $\hat{\beta}_0$ and $\hat{\beta}_1$ are the best linear unbiased estimators (BLUE) of β_0 and β_1 , respectively.**

The Gauss-Markov Theorem states that among all the linear unbiased estimators of β_0 and β_1 , the OLS estimator has the smallest variance (the most efficient).

That is, for any alternative estimators $\tilde{\beta}_0$ and $\tilde{\beta}_1$ that are linear and unbiased, we have

$$Var[\hat{\beta}_0] \leq Var[\tilde{\beta}_0], \quad Var[\hat{\beta}_1] \leq Var[\tilde{\beta}_1].$$

Therefore, when the OLS assumptions hold, we need not look for alternative linear unbiased estimator: None will be better than OLS.

2. (a)

$$E[\bar{Y}_1] = E\left[\frac{1}{N_1} \sum_{i=1}^{N_1} Y_{1i}\right] = \frac{1}{N_1} \sum_{i=1}^{N_1} E[Y_{1i}] = \frac{1}{N_1} \cdot N_1 \cdot \mu_1 = \mu_1. \quad (4)$$

Similarly, we have $E[\bar{Y}_2] = \mu_2$.

$$\begin{aligned} Var[\bar{Y}_1] &= Var\left[\frac{1}{N_1} \sum_{i=1}^{N_1} Y_{1i}\right] = \frac{1}{N_1^2} Var\left[\sum_{i=1}^{N_1} Y_{1i}\right] \\ &= \frac{1}{N_1^2} (Var[Y_{11}] + Var[Y_{12}] + \dots + Var[Y_{1N_1}]) \\ &= \frac{1}{N_1^2} \cdot N_1 \cdot \sigma_1^2 = \frac{\sigma_1^2}{N_1}, \end{aligned} \quad (5)$$

where the third equality follows from the fact that the observations are **independent from each other**, and the fourth equality follows from the fact that the observations are **identically distributed** so that they have all the same variances.

Similarly, we have $Var[\bar{Y}_2] = \sigma_2^2/N_2$.

(b) $E[\hat{d}] = E[\bar{Y}_1 - \bar{Y}_2] = E[\bar{Y}_1] - E[\bar{Y}_2] = \mu_1 - \mu_2$. This implies that $\hat{d}$ is an unbiased estimator for the parameter of interest.

$Cov[\bar{Y}_1, \bar{Y}_2] = 0$ because the SG and HK samples are independent from each other.

By using the formula that $Var[X - Y] = Var[X] + Var[Y] - 2Cov[X, Y]$, where X and Y are two arbitrary random variables, we obtain

$$\begin{aligned} Var[\hat{d}] &= Var[\bar{Y}_1 - \bar{Y}_2] = Var[\bar{Y}_1] + Var[\bar{Y}_2] - 2Cov[\bar{Y}_1, \bar{Y}_2] \\ &= Var[\bar{Y}_1] + Var[\bar{Y}_2] = \frac{\sigma_1^2}{N_1} + \frac{\sigma_2^2}{N_2}. \end{aligned} \quad (6)$$

3. (a) This follows directly from the second equation of the FOC for the OLS estimators:

$$\sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) X_i = 0, \quad (7)$$

i.e., $\sum_{i=1}^n \hat{u}_i X_i = 0$.

(b) We have

$$\sum_{i=1}^n \hat{u}_i \hat{Y}_i = \sum_{i=1}^n \hat{u}_i (\hat{\beta}_0 + \hat{\beta}_1 X_i) \quad (8)$$

$$= \sum_{i=1}^n \hat{\beta}_0 \hat{u}_i + \sum_{i=1}^n \hat{\beta}_1 \hat{u}_i X_i \quad (9)$$

$$= \hat{\beta}_0 \sum_{i=1}^n \hat{u}_i + \hat{\beta}_1 \sum_{i=1}^n \hat{u}_i X_i \quad (10)$$

$$= 0, \quad (11)$$

where the last equality follows from $\sum_{i=1}^n \hat{u}_i = 0$ and $\sum_{i=1}^n \hat{u}_i X_i = 0$.

(c) We have

$$TSS = \sum_{i=1}^n (Y_i - \bar{Y})^2 \quad (12)$$

$$= \sum_{i=1}^n [(Y_i - \hat{Y}_i) + (\hat{Y}_i - \bar{Y})]^2 \quad (13)$$

$$= \sum_{i=1}^n [\hat{u}_i + (\hat{Y}_i - \bar{Y})]^2 \quad (14)$$

$$= \sum_{i=1}^n \hat{u}_i^2 + \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2 + \sum_{i=1}^n 2\hat{u}_i(\hat{Y}_i - \bar{Y}) \quad (15)$$

$$= SSR + ESS, \quad (16)$$

where the last equality follows from $\sum_{i=1}^n 2\hat{u}_i(\hat{Y}_i - \bar{Y}) = 2\sum_{i=1}^n \hat{u}_i(\hat{Y}_i - \bar{Y}) = 2(\sum_{i=1}^n \hat{u}_i\hat{Y}_i - \bar{Y}\sum_{i=1}^n \hat{u}_i) = 0$, by using $\sum_{i=1}^n \hat{u}_i\hat{Y}_i = 0$ and $\sum_{i=1}^n \hat{u}_i = 0$.

Note: This result implies that

$$R^2 = \frac{ESS}{TSS} = \frac{ESS}{ESS + SSR}. \quad (17)$$

As $SSR = \sum_{i=1}^n \hat{u}_i^2$ cannot be negative, we always have $ESS \leq TSS$ and thus R^2 cannot be larger than 1.

(d) As discussed in class, the first equation of the FOC of the OLS estimators $\sum_{i=1}^n \hat{u}_i = 0$ is equivalent to

$$\frac{1}{n} \sum_{i=1}^n \hat{u}_i = 0, \quad (18)$$

that is, the sample average of the OLS residuals is equal to zero. This is designed to mimic:

$$E[u_i] = 0, \quad (19)$$

that is, the expected value (population mean) of the error terms is equal to zero: the sample average mimics the population mean, and the residuals mimic the error terms.

Similarly, we note that the second equation of the FOC $\sum_{i=1}^n \hat{u}_i X_i = 0$ is equivalent to

$$\frac{1}{n} \sum_{i=1}^n \hat{u}_i X_i = 0, \quad (20)$$

that is, the sample average of $\hat{u}_i X_i$ is equal to zero. This is designed to mimic:

$$E[u_i X_i] = 0, \quad (21)$$

that is, the expected value of $u_i X_i$ is equal to zero. Note that (21) is useful because together with (19), it implies that $Cov(u_i, X_i) = 0$ (equivalently, $corr(u_i, X_i) = 0$, the error term u_i is uncorrelated with the regressor X_i):

$$Cov(u_i, X_i) = E[u_i X_i] - E[u_i]E[X_i] = 0. \quad (22)$$

This is important because if u_i and X_i are correlated with each other, then OLS Assumption 1 can not hold (e.g., if the value of u_i tends to increase/decrease when X_i increases/decreases, then OLS Assumption 1 is violated).

4. (a) The dummy variables are defined as follows:

$top10_i$ is equal to one if the i -th observation is among the Top 10 schools, and is equal to zero otherwise.

$r11_25_i$ is equal to one if the i -th observation is among the 11th-25th schools, and is equal to zero otherwise.

$r26_50_i$ is equal to one if the i -th observation is among the 26th-50th schools, and is equal to zero otherwise.

Here, there are totally 4 categories, so we just need 3 dummy variables.

- (b) The linear regression models can be written as follows:

For Top 10 schools: $salary_i = (\beta_0 + \beta_1) + \beta_4 GPA_i + u_i$,

For 11th-25th schools: $salary_i = (\beta_0 + \beta_2) + \beta_4 GPA_i + u_i$,

For 26th-50th schools: $salary_i = (\beta_0 + \beta_3) + \beta_4 GPA_i + u_i$,

For below 50th schools: $salary_i = \beta_0 + \beta_4 GPA_i + u_i$.

Notice that for the case with Top10, 11th-25th, or 26th-50th schools, only one dummy variable will be switched on (equal to one), while the others will be switched off (equal to zero). For the case with below 50th schools (based group), all the dummy variables will be switched off.

- (c) Refer to the figure below.

